Paleomagnetism and the origin of the Mackenzie Arc of northwestern Canada[1]

JOHN K. PARK
Continental Geoscience Division, Geological Survey of Canada, 601 Booth St., Ottawa, Ont., Canada K1A 0E8
DONALD K. NORRIS
RR #4, 444 Okaview Rd., Kelowna, B.C., Canada V1Y 7R3
AND
ANDRÉ LAROCHELLE[2]

Received November 14, 1988
Revision accepted March 29, 1989

Analysis of paleomagnetic data obtained from 1966 alternating-field treatment and from recent thermal demagnetization of the same samples of Late Proterozoic (770 Ma) diabase sills and dykes distributed about the Mackenzie Arc from northeastern British Columbia to the Alaskan border has revealed a primary magnetization in seven sites that is similar to existing data from 10 sites confined to the central Mackenzie Mountains region (N = 17 site poles; 222.2°W, 01.6°N; R = 16.73; K = 60; A_{95} = 5°). The diabases are confined to the dominantly clastic Late Proterozoic Tsezotene Formation and Katherine Group of the Mackenzie Mountains Supergroup. Tests of Carey's orocline hypothesis for the arc using linear regression and a plan-view application of the fold test suggest, in line with earlier structural studies, that the arc is largely nonrotational and that it is not an orocline resulting from the Cretaceous and early Tertiary Laramide Orogeny. Rather, it conforms to the arcuate foreland margin predating deposition of the Late Proterozoic Mackenzie Mountains Supergroup.

L'étude des données paléomagnétiques obtenues, en 1966, au moyen d'aisaimantation par champ alternatif et, récemment, par traitement thermique des mêmes échantillons de filons-couches et de dykes de diabase, d'âge protérozoïque tardif (770 Ma), distribués aux environs de l'arche Mackenzie, du nord-est de la Colombie-Britannique jusqu'à la frontière de l'Alaska, a indiqué une aimantation primaire dans sept sites et dont les résultats sont similaires à ceux déjà rapportés pour 10 autres sites de l'intérieur de la région centrale des monts Mackenzie (N = 17 sites de pôles; 222,2°W, 01,6°N; R = 16,73; K = 60; A_{95} = 5°). Les diabases sont confinées aux roches principalement clastiques, dans la Formation de Tsezotene et dans le Groupe Katherine du Supergroupe de Mackenzie Mountains, d'âge protérozoïque tardif. Les tests appliqués à l'arche de l'hypothèse oroclinale de Carey, au moyen d'un calcul de régression linéaire et une application de vue en plan du test de plissement, suggèrent et corroborent les études structurales antérieures, que l'arche est principalement nonrotationnelle et qu'elle diffère d'un oroclinal qui aurait pu être créé par l'orogenèse de Laramide entre le Crétacé et le Tertiaire précoce. Elle correspond plutôt à la marge arquée d'un avant-pays, plus ancienne que le dépôt du Supergroupe de Mackenzie Mountains au Protérozoïque tardif.

[Traduit par la revue]

Can. J. Earth Sci. **26**, 2194–2203 (1989)

Introduction

Curved orogenic belts have been explained as either primary (Wilson 1954) or secondary (Carey 1958) megascopic structures, inheriting their curvature either from the original shape of the craton margin or from later dynamic bending of the margin. Carey (p. 191) defined curved orogenic belts as *oroclines*, that is, "orogenic belt[s] with a change in trend which is interpreted as an impressed strain." He suggested that smaller oroclines arose by late-stage deformation of geosynclines, whereas large oroclines formed in response to continental drift (shifting tectonic plates). Marshak (1988) used the terms *rotational arc* for Carey's orocline and *nonrotational arc* for "[a] bent [orogen] in which segments of the orogen do not change strike during development of the bend."

Whether the curvature of orogenic belts is primary or secondary may be difficult to determine for many reasons (Marshak 1988), not the least of which is the superimposition of secondary tectonic effects such as rigid block rotations. The existence of oroclinal deformation may be indicated in general by systematic rotation of magnetization declinations along a structural bend, as first suggested by Norris and Black (1961). However, it may not always be possible on the basis of available declination information to distinguish between rotational (oroclinal) and nonrotational arcs because some patterns of deformation affecting both types of arcs may be similar (Marshak 1988). In addition, because rotation may involve blocks of rock of differing scale, it may not be possible without additional geological or geophysical information to tell whether rotation resulted from local deformation of a rock outcrop or from rigid rotation, for example, of a thrust sheet. Bending by both plastic deformation (e.g., Bazhenov and Burtman 1982, 1986) and thrust-sheet rotation (e.g., Kotásek and Krs 1965; Grubbs and Van der Voo 1976; Schwartz and Van der Voo 1984) has been reported. In order to distinguish between rotational and nonrotational arcs and, in turn, to document their deformational history, not only magnetization declination data but also detailed data on directions of tectonic transport, structural geometry, and distribution of strain (see Marshak 1988) are needed.

Paleomagnetic analysis of bends involves at least two problems. First, it is not possible using paleomagnetic methods to distinguish between arcs with primary curvature and those with secondary curvature produced by differential transport of fault-bounded blocks without rotation (Knowles and Opdyke 1968; Marshak 1988). Second, one cannot distinguish between bends formed coevally with the magnetization and those formed later (Marshak 1988).

[1]Geological Survey of Canada Contribution 39187.
[2]Deceased, formerly with the Geological Survey of Canada.

Printed in Canada / Imprimé au Canada

Paleomagnetic studies suggest that arcuate segments outlining the eastern margin of the North American Cordillera reflect the shape of the foreland margin (Norris and Black 1962; Grubbs and Van der Voo 1976; Schwartz and Van der Voo 1984). In Canada, the 150 km long Crowsnest Deflection was shown to be a nonrotational arc from studies of abundant structural, stratigraphic, kinematic, and paleomagnetic data (Norris and Black 1962; Norris 1968).

From a structural study of en échelon folding and crustal shortening farther north in the Mackenzie Arc, Norris (1972) concluded that, similar to the Crowsnest Deflection, the curvature was primarily due to the initial geometry of the margin of the miogeocline. The arc, along with its complementary re-entrants, was inherited from the ancient, persistent curvilinear trace of the eastern margin of the miogeocline, which in turn controlled the ultimate curvilinear shape of the Foreland Thrust and Fold Belt. Isopach data from three widely spaced late Precambrian formations from the Mackenzie Mountains tend to support this conclusion (Aitken and Long 1978). Although structural bending would naturally produce arcuate isopachs, Aitken and Long (p. 628) pointed out that reconnaissance geological mapping had not provided evidence of large-scale extension parallel to fold axes, at least in the frontal folds of the arc.

Norris and Larochelle in 1966 attempted to use paleomagnetism to test whether the Mackenzie Arc was an orocline by studying Late Proterozoic (770 Ma; Armstrong *et al.* 1982) diabase sills and dykes from around the arc. However, the then-current laboratory methods and the absence of comparative paleomagnetic data from the region precluded interpretation of the results. The sample collection and initial data were later given to the senior author, and after additional experiments were performed, notably thermal demagnetization, the results were combined with those from another paleomagnetic study of Mackenzie Mountains diabases (Park 1981). Collectively, the data imply that the arcuate shape of the orogenic belt is essentially, if not wholly, primary in nature. It is not orocline.

Geological setting of the Mackenzie Arc

Along the flank of the northern Cordillera in western District of Mackenzie and eastern Yukon Territory, the structure of the predominantly sedimentary supracrustal wedge is dominated by the great arc embracing the Canyon and Backbone ranges of the Mackenzie Foldbelt (Norris and Yorath 1981). The arc, extending from the Liard River just south of the 60th parallel in British Columbia to the Knorr Range just north of the 65th parallel in eastern Yukon Territory, has a length of about 950 km (Fig. 1). Over much of its length the front of the arc is delineated by upturned or contraction-faulted Lower and Middle Devonian carbonate and clastic rocks. Immediately to the east is the parautochthon of the Mackenzie Plain and Franklin Mountains, constituting the western margin of the Interior Platform. To the west is the Selwyn Foldbelt, dominated by Proterozoic through middle Paleozoic sedimentary rocks with associated mid-Cretaceous silicic intrusions.

The stratigraphic succession within the Mackenzie Arc is characterized by shelf and miogeoclinal sedimentary rocks. Deposition of this sequence was punctuated by orogenic events in the late Helikian, late Hadrynian, mid-Cretaceous, and Late Cretaceous to early Tertiary. Additionally, epeirogenic events created significant disconformities in the mid-Cambrian,

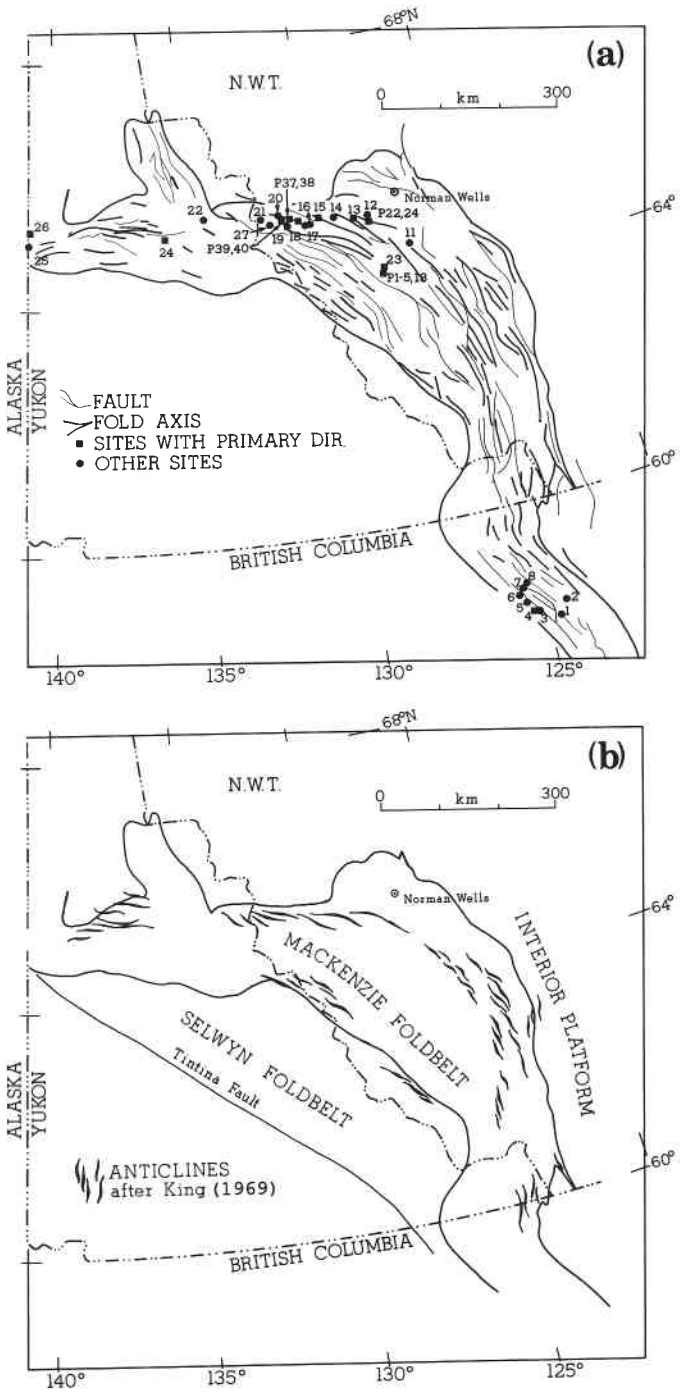


FIG. 1. (*a*) Tectonic map of the Mackenzie Arc (after Stockwell 1969), showing paleomagnetic sampling sites in diabase. Sites prefaced by "P" are from Park (1981). (*b*) The Mackenzie Arc showing traces of en échelon folds (after Norris 1972).

mid-Ordovician, mid-Silurian, early Late Devonian, and Early Cretaceous. Mafic sills and dykes intrude quartzites and shales of the Middle Proterozoic Tsezotene Formation and Katherine Group and have been dated at about 770 Ma (Armstrong *et al.* 1982). Because the sills are layer parallel, simple back-tilting of bedding into the horizontal about the strike was the only local structural correction applied to data from both sills and dykes.

In contrast to some parts of the Foreland Thrust and Fold Belt, the Mackenzie Arc is characterized by bundle of flexural-

slip, cylindrical folds. Most folds have curviplanar axial surfaces dipping steeply toward or away from the craton. Some folds, however, are symmetrical, whereas others reverse their direction of vergence along strike.

Individual folds or fold pairs are characteristically en échelon, so that they lie systematically to the right or left of one another, depending upon their position along the Mackenzie Arc (Norris 1972) (Fig. 1*b*). On the north-trending leg of the arc the folds are preferentially right-hand en échelon, and they are associated with contraction-faulted mountian ranges whose trends can be strongly curvilinear. On the outermost part of the arc, opposite Norman Wells, both right- and left-hand configurations of fold axes are equally common; on the west-trending leg of the arc, folds are characteristically left-hand en échelon, that is, the reverse of the configuration on the north-trending leg. If this regional folding pattern were due to bending about a vertical axis, the arrays on the north-trending leg of the Mackenzie Arc would have been left-hand ones and on the west-trending leg, right-hand ones, exactly the opposite of what is observed.

Structural arguments, therefore, strongly support the hypothesis that the Mackenzie Arc is simply a reflection of the original shape of the eastern margin of the miogeocline.

Methods

Samples were collected at 26 sites (Fig. 1) in 1966 by Larochelle and Norris with a portable drill and 3.18 cm (1.25 in.) bits. Generally, 5–10 samples were drilled at each site and oriented by magnetic compass. In the laboratory, samples were cut into two specimens 2.7 cm in length.

Natural remanent magnetizations (NRM) were measured on a biastatic magnetometer (Larochelle and Christie 1967). Specimens were demagnetized on an in-house apparatus (Larochelle and Black 1965). One test specimen from each site was treated in peak alternating fields (AF) of 10, 20, 25, 30, and 50 mT; the remaining specimens were treated in fields of 5, 10, and 20 mT. Site 27 in the Mackenzie Mountains, 20 km east of Arctic Red River, was drilled with a 2.54 cm (1 in.) bit, and AF demagnetization was done separately.

Specimens given to Park in 1984 for further study were pared down to current standard specimen size (about 2.5 cm diameter and 2.2 cm height) and were then remeasured with either Schonstedt (DSM-1) or Geofyzika (JR-4) spinner magnetometers. Comparison of Larochelle and Park measurements was facilitated by a set of international test specimens circulated by D. Collinson in 1974. A previous study of related diabases in the Mackenzie Mountains (Park 1981) revealed that the most effective method for analyzing the NRM was the use of AF treatment followed by thermal demagnetization. Therefore, after processing all specimens at the previous 20 mT peak field, we carried out thermal treatments in steps from 100°C to as high as 650°C, using in-house instruments (Roy *et al.* 1972, 1973) and a Schonstedt TSD-1 furnace. Ambient fileds in the in-house instruments were typically less than 2 nT. Small, portable magnetic shields, with internal fields of less than 10 nT, were used to transport specimens between instruments.

We determined magnetization directions from vector plots by drawing straight lines by eye through apparent straight-line segments.

Site-mean directions were converted to virtual paleomagnetic poles for analyses used to determine whether the Mackenzie Arc is nonrotational. Poles were used for two reasons. First, the distribution of localities over a distance of about 600 km would mean that magnetization directions would not remain parallel or be directly correlated with the changing regional strike, as required for the test. Second, the angular distance (ancient colatitude) between poles and localities is almost 90°; therefore any changes in declination would correlate almost directly with changes in regional strike.

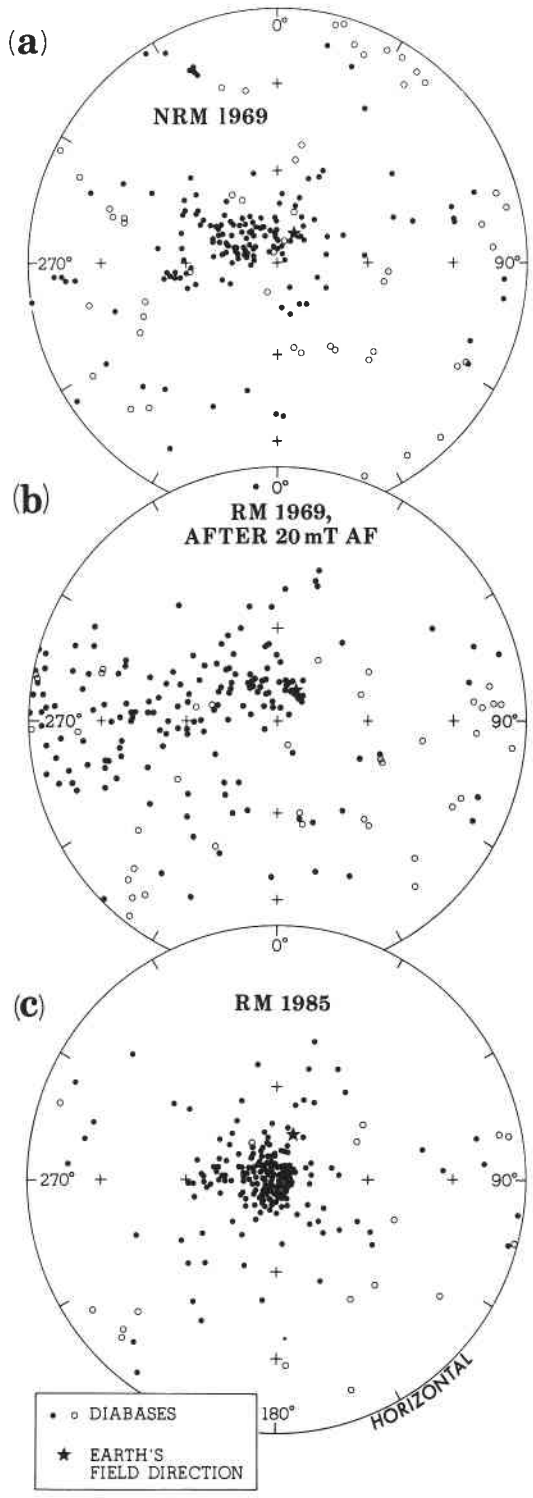


FIG. 2. Natural remanent magnetization (NRM) directions of specimens and later remanent magnetization (RM) acquired during storage. Closed symbols represent downward-directed magnetizations; open symbols, upward-directed magnetizations.

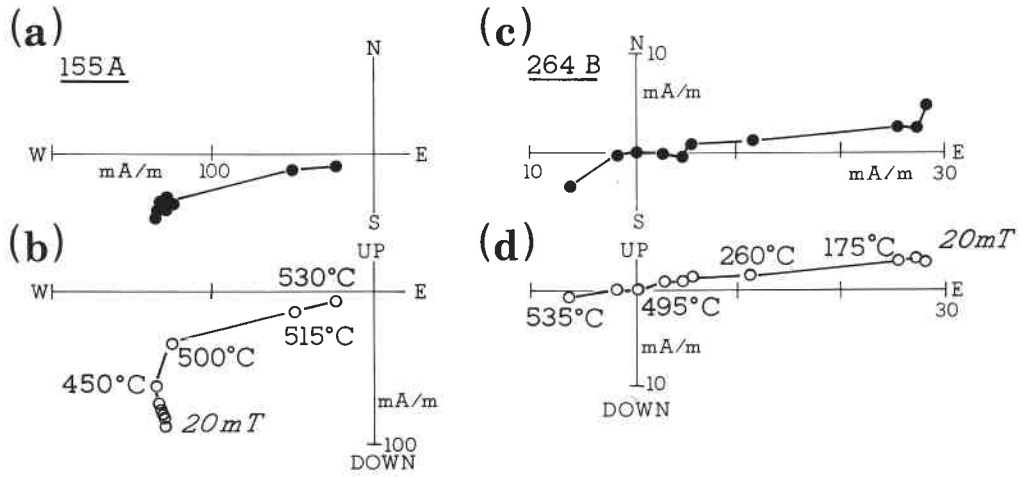


FIG. 3. Vector diagrams showing examples from sites 15 and 26 of the decay of magnetization with thermal treatment in the horizontal (N−E) and vertical planes (UP−E). Data represent *in situ* directions.

Three methods were used for the orocline test: linear regression (Bazhenov 1982; Schwartz and Van der Voo 1983; Eldredge *et al.* 1985) and two versions of the fold test (McElhinny 1964; McFadden and Jones 1981). In the first method, magnetization declinations (or, in our case, longitudes of paleomagnetic poles) were plotted against regional strikes on a simple orthogonal plot. If the arc resulted from tectonic bending, the angle between polar longitude and regional strike would be generally constant. The best-fit line through the data was compared with a line of unit positive slope, corresponding to ideal oroclinal bending; that is, the change in strike would correspond to a change in magnetization declination.

For the second method, localities about the arc were arbitrarily divided into two groups, as if they were on opposite limbs of a fold, with structural trends treated like the bedding dips of structural cross sections. After testing that the precisions of poles from the two "limbs" were the same, we compared calculated and tabled values of a statistic to determine whether the two groups had the same mean paleomagnetic pole (McFadden and Lowes 1981; McFadden and Jones 1981). The null hypothesis, that the two groups have a common true mean pole, is rejected at the level of significance chosen if the calculated value is greater than the tabled value.

For the third method, Fisher's (1953) precision parameter (k) was compared before and after an assumed correction for structural bending had been made, and a variance ratio of the precisions was compared with tabled F values (McElhinny 1964). This test is not strictly valid; however, it will confirm conclusions drawn using the second method, because the conditions of its application are more stringent (McFadden and Jones 1981). It is used in addition to the second method because it at least indicates the lowest level at which the test is significant and does not rely on an arbitrary separation of the data into two groups.

Results and analysis

Resolution of natural remanent magnetization

NRM directions measured in 1969 by Larochelle are shown in Fig. 2*a*. Directions are scattered, but they cluster away from the present Earth's field (PEF) direction with steep positive inclinations in the northwest quadrant.

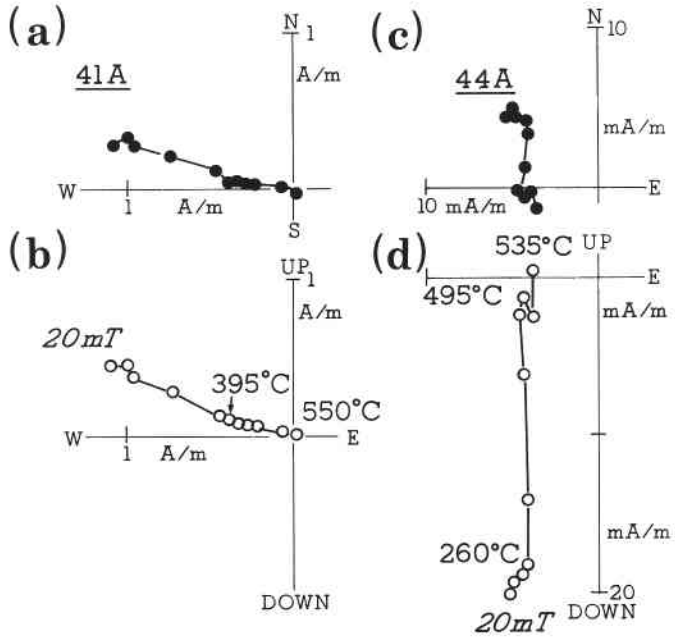


FIG. 4. Vector diagrams showing examples of the decay of magnetization in the horizontal (N−E) and vertical planes (UP−E) in two specimens from site 4 subjected to thermal treatment. Data represent *in situ* directions. Specimen 44A acquired a large induced magnetization at temperatures above 535°C. Differing magnetic behaviour of 41A and 44A may reflect different positions within the dyke.

With AF treatment at 20 mT the directions of remanent magnetization tend to become distributed between the PEF direction and a horizontal westerly direction (Fig. 2*b*). The NRM intensity decreased by factors of 2.5 to 67, with an average decrease of about 14; the large intensity decrease and a sometimes erratic decay indicate the removal of a largely unstable magnetization.

Specimen directions, as remeasured by Park in 1985 prior to treatment, are shown in Fig. 2*c*. Directions are tightly clustered about the vertical owing to the acquisition of a viscous remanent magnetization (VRM) during 17 years of storage. This VRM was a major obstacle to resolving the magnetization components present.

TABLE 1. Paleomagnetic data obtained by thermal and AF[a] treatment

Site	Unit	T_{UB} (°C)	N (n)	Direction (*in situ*) D (°), I (°)	R	k	α_{95} (°)	Pole (°W, °N)	Direction (bedding) D (°), I (°)	Pole (°W, °N)
1	Dyke	100–400	4 (6)	*232, +78*	*3.90*	*30*	*17*	*149, +40*	*039, +81*	*094, +69*
3	Sill	200–350	1 (9)	317, +57	—	—	—	—	—	—
4	Dyke	260–495	5	*346, +84*	*4.98*	*172*	*6*	*134, +70*	*268, +56*	*193, +30*
		330–>535	4	275, +06	3.85	20	21	218, +05	277, −25	228, −08
6	Dyke	520–550	7 (1)	*029, +82*	*6.73*	*22*	*13*	*101, +71*	*332, +68*	*229, +72*
8	Dyke	260–550	9	*344, +78*	*8.83*	*46*	*8*	*162, +79*	*317, +74*	*194, +68*
11	Dyke	175–535	8 (3)	*283, +86*	*7.81*	*36*	*9*	*145, +65*	*081, +80*	*084, +61*
13[a]	Sill	—	8	276, +06	7.83	42	9	223, +05	276, +06	223, +05
15[a]	Sill	—	8	258, +08	7.78	31	10	207, −01	258, +08	207, −01
18[a]	Sill	—	10	259, +25	9.87	67	6	205, +07	259, +25	205, +07
23[a]	Sill	—	10	279, +07	9.86	66	6	225, +07	279, +08	224, +08
24[b]	Dyke	330–535	7 (3)	100, −07	—	430	8	234, +08	100, −07	234, +08
26	Dyke	395–>515	6 (5)	094, −31	5.86	36	11	228, +17	100, +02	240, +03

NOTES: T_{UB}, unblocking-temperature range of magnetization component; N (n), number of specimens yielding (not yielding) component; D, I, declination and inclination of direction, positive downward; R, vector resultant; k, Fisher's index of precision; α_{95}, 95% confidence error about the mean direction. Sites not included: 2, 7, 17 (probable lightning strike); 5, 22, 25 (no useful data); 27 (mixed components); 12, 14, 16, 19, 20, 21 (recent viscous remanent magnetization). Laramide component in italics.

[a]Results calculated from unpublished 1969 data of Larochelle.

[b]Average direction of site 24 is calculated from converging remagnetization circles (Halls 1976; Bailey and Halls 1984).

TABLE 2. Averages of diabase site poles

Magnetization	N	*In situ* Pole (°W, °N)	R	K_2	A_{95} (°)	Tilt-corrected Pole (°W, °N)	R	K_1	A_{95} (°)	K_2/K_1
S_C	5	*139.8, +66.4*	*4.83*	*23.08*	*16*	166.2, +71.2	4.41	6.83	32	3.38
S_A[a]	17	220.8, +04.8	16.75	64.55	5	*222.2, +01.6*	*16.73*	*59.76*	*5*	~1
S_A[b]	15	219.5, +03.9	14.81	74.77	5	220.2, +01.0	14.82	76.41	5	~1

NOTES: N is the number of sites; R is the vector resultant of the site poles; K is the precision of the poles; A_{95} is the error about the mean pole; K_2/K_1 is the variance ratio (F-distribution). Data in italics are considered as representing the best summaries of the results.

[a]Includes all sites given in Table 3 and site 4 from Table 1.

[b]Excludes the farthest western sites, 24 and 26.

Subsequent thermal demagnetization after AF treatment in 20 mT revealed both a shallowly inclined magnetization and a steeply inclined magnetization in the diabases. The steeply inclined magnetization may comprise three different components, two of which probably relate to the storage history in Calgary and in Ottawa. The earlier storage component acquired in Calgary has unblocking temperatures (T_{UB}'s) up to about 500°C (Figs. 3*a*, 3*b*), which are also typical of the third steeply inclined component (Figs. 4*c*, 4*d*). Isolation of the third component (S_C) from the two storage components was only achieved where there was a clear separation of T_{UB} ranges, principally within sites from northeastern British Columbia (1, 3, 4, 6, 8) (Table 1).

The second well-defined magnetization component, herein called S_A, has a shallowly inclined easterly or westerly direction (Figs. 3, 4). At four sites (13, 15, 18, 23) S_A is defined from AF data from Larochelle's original study (Table 1). Site 26 records antiparallel magnetization components, as shown by reflection of the directions through the origin (Figs. 3*c*, 3*d*). Data from this site and several others suggest the presence of two magnetic components with similar directions below 535°C (e.g., site 4, Figs. 4*a*, 4*b*; site 26, Figs. 3*c*, 3*d*); these are discussed below. Site directions are given in Table 1.

Tilt test

The fold test of McFadden and Jones (1981) is inappropriate for this study because most sites are located on separate folds; therefore, the fold test of McElhinny (1964) was used. The five S_C site poles (1, 4, 6, 8, 11) (Table 1) were averaged with and without bedding correction (Table 2). The calculated variance ratio (K_2/K_1) of 3.38 lies between F variates of 3.44 and 2.59 for the 95 and 90% confidence limits, respectively, indicating that the tilt test is only significant at the 90% level. This suggests that S_C is post folding. For a similar tilt test of S_A, the seven poles of this study (Table 1) were combined with the 10 poles derived from the results of Park (1981) (Table 3). We recalculated Park's results by combining his A_R and B directions, giving individual specimen directions unit weight (see Tables 2 and 3 of Park 1981). The variance ratio is almost unity, and the test is therefore inconclusive (Table 2). However, other considerations, principally arguments establishing the primary nature of the magnetization (Park 1981), strongly suggest that S_A is prefolding.

Magnetic mineralogy

The magnetic mineralogy of diabases in this study is very similar to that described by Park (1981). The major magnetic

TABLE 3. Site pole directions uncorrected and corrected for assumed structural bending

			No correction assumed					Correction for bending assumed		
Locality	Site(s)	N	D (°), I (°)	R	k	Pole (°W, °N)	δp (°), δm (°)	Δ (°)	D' (°), I (°)	Pole (°W, °N)
	4	4	277, −25	3.85	20	228, −08	12, 23			
	10	14	280, +15	13.56	29	223, +11	4, 8			
1	P1	1	269, +10	—	—	(215, +04)	—			
1	P2	8	270, +06	7.75	28	217, −03	5, 11	0	270, +06	217, −03
1	P3	2	271, −03	1.99	—	220, −01	—	0	271, −03	220, −01
1	P4	6	278, +12	5.94	85	222, +09	4, 7	0	278, +12	222, +09
1	P5	5	270, −12	4.75	16	220, −06	10, 20	0	270, −12	220, −06
1	P13	9	270, +04	8.71	28	217, +02	5, 10	0	270, +04	217, +02
1	23	10	279, +08	9.86	66	224, +08	3, 6	0	279, +08	224, +08
1	*P2−P13, 23*	*6*	*273, +03*	*5.93*	*71*	*220, +02*	*4, 8*	*0*	*273, +03*	*220, +03*
2	P22	2	280, +11	1.96	—	225, +09	—	9	289, +11	233, +13
2	P24	7	273, +01	6.76	25	221, +02	6, 12	9	282, +11	229, +05
2	*P22, P24*	*2*	*276, +06*	*1.99*	—	*223, +06*	—	*9*	*285, +06*	*230, +09*
3	*13*	*8*	*276, +06*	*7.83*	*42*	*223, +05*	*4, 9*	*8*	*284, +06*	*230, +09*
4	*15*	*8*	*258, +08*	*7.78*	*31*	*207, −01*	*5, 10*	*18*	*276, +08*	*224, +06*
5	*18*	*10*	*259, +25*	*9.87*	*67*	*205, +07*	*3, 6*	*20*	*279, +25*	*223, +16*
6	P37	1	270, −16	—	—	(224, −07)	—	—	—	—
6	*P38*	*4*	*276, −07*	*3.44*	*5*	*228, −01*	*22, 42*	*26*	*302, −07*	*252, +10*
7	P39	5	277, −16	4.69	13	231, −04	12, 23	27	304, −16	256, +06
7	P40	6	262, −08	5.49	10	215, −07	11, 23	27	289, −08	240, +04
7	*P39, P40*	*2*	*269, −12*	*1.98*	—	*223, −06*	—	*27*	*296, −12*	*248, +05*
8	*24*	*7*	*280, +07*	—	*430*	*234, +08*	—	*37*	*317, +07*	*269, +21*
9	*26*	*6*	*280, −02*	*5.86*	*36*	*240, +03*	*6, 11*	*82*	*362, −02*	*323, +24*

NOTES: Sites prefaced by "P" are from Park (1981). The directions of the P-site data are averages combining all individual A_R (Park's Table 2) and B directions (Park's Table 3). N is the number of specimens or sites; for P sites, N is the number of directions, usually equal to the number of specimens; Δ is the correction applied to declination D, according to the structural trend at the site; $D' = D + \Delta$. For convenience, sites P2−P13 and 23 (locality 1) are assumed to need no correction. δp, δm are the semi-axes of the ellipse of confidence at $P = 0.05$ about the pole calculated from the average site direction. Italicized values represent data from individual localities used in orocline test. Data from sites with only one specimen (P1, P37) and sites outside the arc (4) are excluded from the test. Other symbols are defined in Table 1.

mineral is magnetite is magnetite with exsolved ilmenite. Ilmenite, occurring as trellis, sandwich, or composite structures or in myrmekitic intergrowths, is generally pristine, but it may be partly or wholly altered to leucoxene in several sites in northwestern British Columbia. In general, even rather fresh looking magnetite has altered to several microcrystalline phases, which are not readily identifiable. In its most altered state magnetite appears to have been either replaced or completely dissolved, leaving a network of ilmenite laths. The microgranular phases of the least-altered grains probably include maghemite and hematite, evidence for which was discussed in Park (1981). Additional opaque minerals include sulfides, commonly associated with the ilmenite−magnetite grains.

Carriers of the magnetization

Primary magnetization resolved in the previous diabase study was thought to occur as both normal and reverse remanences in maghemite (A_R, A_N) and hematite (B_R, B_N) (Park 1981). A third carrier, magnetite, was considered as containing part of an unresolved magnetization R_M, thought to be composed of antiparallel remanences with the A_R-B_R and A_N-B_N directions. In the present study there are normal and reverse remanence directions that have the magnetite blocking-temperature range (Figs. 3*c*, 3*d*, 4*a*, 4*b*), thus supporting the previous conjecture on the nature of much of the mixed R_M magnetization (Park 1981). Magnetizations residing in hematite were not identified; however, changes in magnetic behaviour at about 350°C suggest magnetization residing in maghemite. The antiparallel remanences with similar direction in both studies are referred to as S_A.

The identification of a low-T_{UB} magnetization with maghemite was not proven but is suggestive. Maghemite has a Curie point possibly close to 645°C (Özdemir and Banerjee 1984); however, it inverts to hematite at temperatures between about 300° and 600°C (Özdemir and Dunlop 1988), depending critically on the presence of impurities (Özdemir and Banerjee 1984). Although the inversion of maghemite between 350° and 450°C in this study could explain the magnetic behaviour, alternative arguments involving magnetite could be offered. The effect could be related to the presence of two generations of magnetite with different grain-size ranges (or unblocking-temperature ranges) (Dankers 1978) and (or) to partial oxidation of magnetite during stepwise heating (Dankers 1978). However, these explanations were clearly discounted by Park (1981). It was shown that with low AF treatment (20−45 mT) prior to heating, the lower T_{UB} component was enhanced relative to the higher T_{UB} magnetite component, indicating that the higher T_{UB} remanence was being preferentially removed. Thus, the lower T_{UB} component resided in another magnetic phase; it was tentatively identified as maghemite because of the magnetic characteristics (relatively high intensities and low unblocking-temperature range) and petrographic evidence for low-temperature alteration in magnetite.

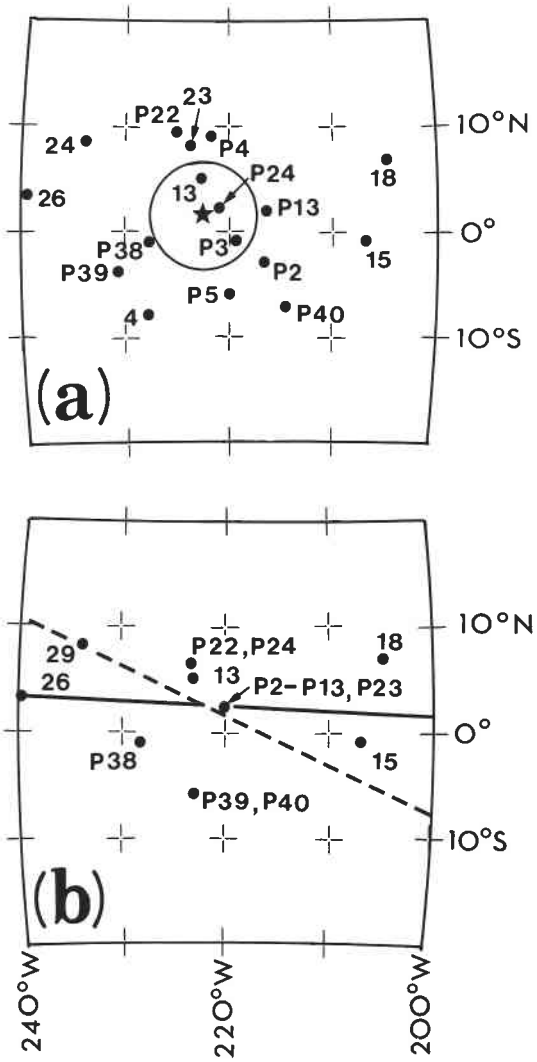


FIG. 5. (*a*) Site poles (17). Mean (star) and pole error (A_{95}) are shown. (*b*) Locality-mean poles (9) after combining site poles at each of three localities. Solid line is the best-fit line through the pole distribution, which has an eccentricity of 0.91 (Engebretson and Beck 1978). Dashed line is a small circle about the mean locality.

The previously postulated seventh magnetization component (Park 1981), forming part of R_M, would coincide with S_C in the present study. It probably was acquired as either a thermal or chemical remanent magnetization during the Laramide Orogeny. A chemical origin is more likely for several reasons. First, S_C is mainly revealed in diabases of northeastern British Columbia where the opaque mineralogy of the diabases is more altered. Some of the affected sites, notably 3 and 4, are near a major fault system that was active during the Laramide Orogeny. Second, samples from site 4 that contain only S_A reveal that the lower T_{UB} range was not affected by heating (Figs. 4*a*, 4*b*), thus generally ruling out a thermal overprint.

In summary, S_A probably resides in magnetite and hematite, and possibly maghemite, as a primary magnetization in the diabases. S_C is likely a post-folding, chemical remanent magnetization, but its host is unknown. Because S_A is primary it is used for the orocline test.

Discussion

The 17 S_A site poles from this and the previous study (Park 1981) are recorded in Table 3. They form a cluster with elongate distribution (Fig. 5*a*). Before we used the data to test the orocline hypothesis, we combined site poles from a single sill at each of three localities (Table 3). This procedure created a more even distribution of localities about the arc and reduced large errors associated with some sites. The resulting distribution of nine locality poles remains elongate (Fig. 5*b*), with an eccentricity of 0.91 (Engebretson and Beck 1978). The orientation of the best-fit line through the distribution is subparallel to a small circle about the mean locality (Fig. 5*b*), at least suggesting that the shape of the distribution results from tectonic rotation. However, because the data base is limited (nine poles), statistical evaluation of the shape of the distribution is not valid. Using the methods described above, one can test the nine locality mean poles to determine whether their distribution relates to the formation of a rotational arc.

Linear regression method

With the first method, pole declinations are plotted against fold-axis trends for each locality about the arc, and the best-fit line through the data is analyzed by linear regression. To produce a suitable graph that passes close to the origin, the actual values plotted are the reference declination (D_r) minus the observed declination (D_o) on the ordinate and the reference strike (S_r) minus the observed strike (S_o) on the abscissa. The reference values are simply means of the respective declination and strike data. Error limits are assigned to polar longitudes according to the formula $D = \sin^{-1}[(\sin A_{95})/(\sin p)]$, where A_{95} is the 95% circle of confidence about the pole and p is the ancient colatitude (angular distance between sampling locality and pole) (Beck 1980).

Declination and strike data are recored in Table 4 and illustrated in Fig. 6. S_o data are derived from geological maps from the averaged values of fold traces (Fig. 1); their determination has estimated errors of $\pm 4°$. Linear-regression analysis produced a line with slope -0.258 ± 0.138 close to the abscissa, away from the line for ideal oroclinal bending, suggesting that the fold belt reflects an original arcuate shape (Eldredge *et al.* 1985), in agreement with the conclusion of Norris (1972) from structural studies. Although the correlation coefficient is low (-0.53) and the χ^2 test indicates that regression is not significant at the 95% level, Student's t test, used for small data sets, indicates that the arcuate shape is not due to oroclinal bending. Using a t statistic of Lowrie and Hirt (1986) to test how the slope of the regression line compares with zero slope, we found the calculated value (1.86) to be less than the tabled value (2.31) at the 95% level.

Fold-test methods

Using the fold test of McFadden and Jones (1981), we divided the nine localities into two groups ($1-7$ and $8 + 9$) and calculated vector resultants and precisions from the data, before and after an assumed correction for bending (see Table 5 for results and application of method). An F test performed on the two groups shows them to have identical precisions at the 97.5% level of the test, both before and after correction for bending, despite changes in precision caused by lack of common "bedding corrections" at the localities within each group. When the fold test is performed, we reject the null hypothesis at the 99% level for poles corrected for bending but accept it for poles not corrected for bending; that is, results strongly suggest that the arc is nonrotational.

Using the fold text of McElhinny (1964), we found the arc to be nonrotational at greater than the 99% significance level.

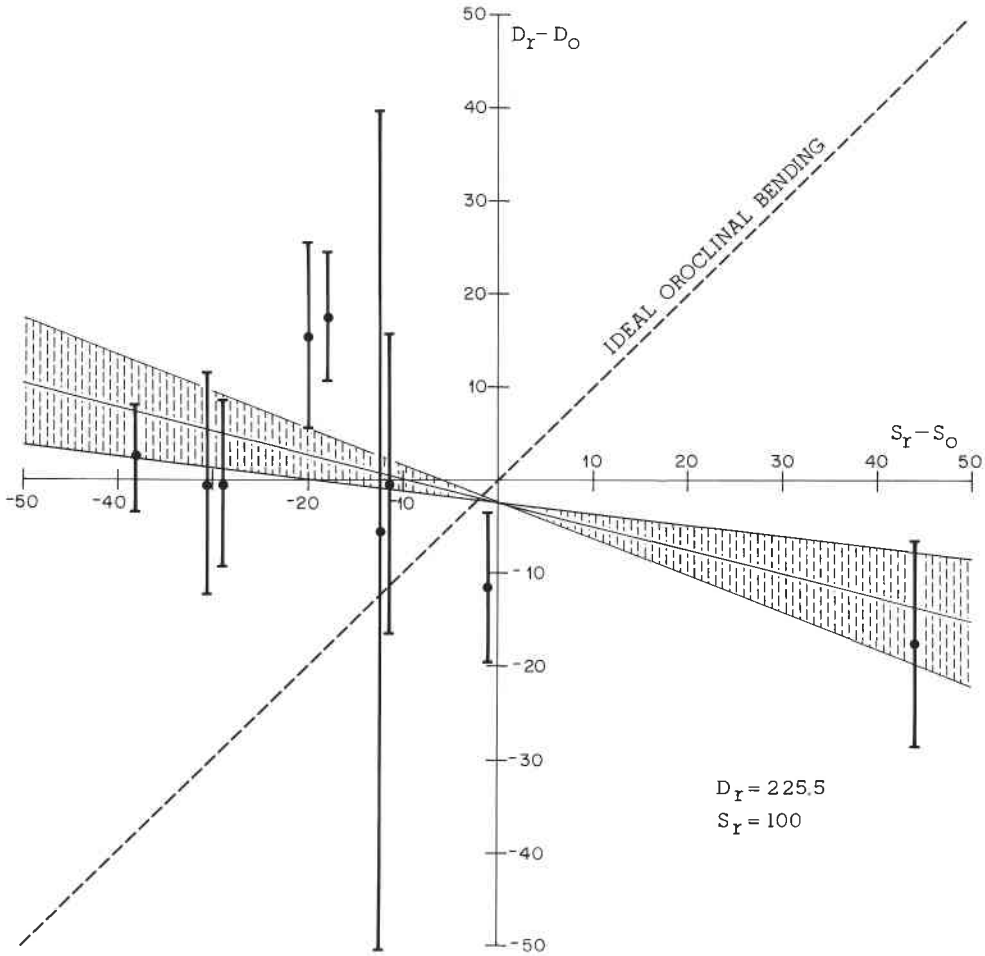


FIG. 6. Plot of declination deviations ($D_r - D_o$) of S_A paleopoles versus strike deviations ($S_r - S_o$) of fold structures from the Mackenzie Arc (see Table 4). D_r and S_r are the reference (mean) declination and strike, respectively, and D_o and S_o are the observed declination and strike, respectively, at each sampling locality. Error bars were calculated after Beck (1980) (see Table 4). The best-fit straight line found by linear regression is characterized by slope -0.258 ± 0.138 and ordinate intercept -2.24 ± 3.97.

TABLE 4. Paleomagnetic data used for orocline test

Locality	D_o (°)	$D_r - D_o$ (°)	A_{95} (°)	P (°)	ΔD (°)	S_o (°)	$S_r - S_o$ (°)
1	220	+2.5	6	89	4	138	−38
2	223	−0.5	12	87	12	129	−29
3	223	−0.5	9	87	9	130	−30
4	207	+15.5	10	85	10	120	−20
5	205	+17.5	7	77	7	118	−18
6	228	−5.5	44	98	45	112	−12
7	223	−0.5	16	96	16	111	−11
8	234	−11.5	8	90	8	101	−1
9	240	−17.5	11	95	11	56	+44

NOTES: D_o is the mean longitude of the paleopole; D_r is the reference longitude (222.5°); S_o is the best mean-strike determination (±3.5°) of fold traces near the locality; S_r is the reference strike (100°); ΔD is the error in longitude, $\sin^{-1}$ [(sin A_{95})/(sin p)], after Beck (1980), where A_{95} is the paleopole error and p is the ancient colatitude (angular distance between paleopole and locality). Values of A_{95}, except that for locality 1, are estimates from the α_{95} errors. At localities 2 and 7 the overall α_{95} error is reduced by the rule for combining errors.

The test, though not strictly valid (McFadden and Jones 1981), indicates that the lowest level of significance is at least 99%.

Comparison of methods

Linear-regression analysis does not indicate a strong correlation of data, but the resultant best-fit line appears to depart significantly from the line that would indicate a rotational arc. Fold-test methods strongly indicate that the Mackenzie Arc is nonrotational. The low significance of results when linear regression is used possibly relates to the uneven distribution of

TABLE 5. "Fold test" for oroclinal bending

Localities	D (°), I (°)	N	R	k	Statistic: Description	Statistic: Value	Significance level (%)
				In situ			
1–7	218.5, +01.9	7	6.9091 (R_a)	65.99 (k_a)	F calculated (k_b/k_a)	3.28	>97.5
8, 9	237.0, +05.5	2	1.9954 (R_b)	216.61 (k_b)	$F_{12, 2}$ from tables, $P = 0.025$	39.41	
1–9		9	8.8209 (R_1)	44.66 (k_1)	S calculated	0.8713	99
					S from tables, $P = 0.025$	0.6941	
					$P = 0.01$	0.9307	
				After unbending			
1–7	232.4, +08.5	7	6.8428 (R_a)	38.17 (k_a)	F calculated (k_a/k_b)	7.07	>97.5
8, 9	295.7, +24.9	2	1.8150 (R_b)	5.40 (k_b)	$F_{12, 2}$ from tables, $P = 0.025$	39.41	
1–9		9	7.8531 (R_2)	6.98 (k_2)	S calculated	2.2423	99
					S from tables, $P = 0.025$	0.6941	
					$P = 0.01$	0.9307	
					$P = 0.0005$	1.9606	
				Comparison			
1–9					F calculated (k_1/k_2)	6.41	>99
					$F_{16, 16}$ from tables, $P = 0.025$	2.77	
					$P = 0.01$	3.37	

NOTES: Localities (see Table 4) are divided into two groups (1–7 and 8+9) so that the fold test of McFadden and Jones (1981) can be used to determine whether the plan-view bending of the Mackenzie Arc is original or was imposed later. Poles rather than directions are used. Two statistics are calculated for the test. First, the F variate is determined to test whether the two groups have a common precision (McFadden and Lowes 1981; McFadden and Jones 1981). Second, a statistic, arbitrarily called S, is calculated for the "fold test" (McFadden and Lowes 1981; McFadden and Jones 1981). Calculated values of $S = [R_a + R_b - R^2/(R_a + R_b)]/[2\ (N - R_a - R_b)]$ are compared with tabled values of $S = (1/P)^{1/(N-2)} - 1$, where R, the magnitude of the resultant vector of all site-mean poles, is R_1 or R_2; R_a and R_b are the magnitudes of the resultant vector of site-mean poles in the two groups; N is the number of localities; and P is the level of significance. The null hypothesis, that the two groups have a common true mean, is rejected if S (calculated) $> S$ (tabled) at the P value chosen. In the bottom part of the table, the fold test of McElhinny (1964) is shown. Although the test is invalid, it almost certainly yields significance levels that are too low because the test is too stringent (McFadden and Jones 1981). Degrees of freedom for all tabled F-values are $2(N - 1)$.

localities. Results from all methods rely heavily on inclusion of the most westerly sites, 24 and 26. If those sites are excluded from the analysis, the best-fit line from linear-regression analysis will shift toward the line for ideal oroclinal bending and the significance level of the fold tests will be reduced to about 90% (Table 5). However, there is no reason for excluding these sites. The low degree of significance in the regression method clearly allows for some modification of a nonrotational arc, as suggested by Norris (1972) from his structural studies.

Additional evidence that the arc is nonrotational is the remarkably good agreement of the paleopole from site 4, well south of the arc (south of Liard River), with the poles from the arc (Table 3).

We conclude that the configuration of the Mackenzie Arc is in large part a reflection of the original geometry of the foreland margin.

Acknowledgments

We thank Gilbert Massie for designing a device to aid in paring down the original specimens to standard size; Benoit St-Louis for specimen preparation; Normand Brunet for some of the paleomagnetic measurements; Charlie Jefferson for supplying a copy of the geochemical work; Donald Harris for microprobe work; Bert Draper for drafting of illustrations; Diane Regimbald for typing; Alf Latham for useful discussions; Walter Fahrig and Ken Buchan for initial reviews of the manuscript; and journal referees for useful comments.

AITKEN, J. D., and LONG, D. G. F. 1978. Mackenzie tectonic arc — reflection of early basin configuration. Geology, **6**: 626–629.

ARMSTRONG, R. L., EISBACHER, G. H., and EVANS, P. D. 1982. Age and stratigraphic–tectonic significance of Proterozoic diabase sheets, Mackenzie Mountains, northwestern Canada. Canadian Journal of Earth Sciences, **19**: 316–323.

BAILEY, R. C., and HALLS, H. C. 1984. Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data. Journal of Geophysics, **54**: 174–182.

BAZHENOV, M. L. 1982. The application of regression analysis in paleomagnetic–tectonic studies. Izvestiya, Academy of Sciences, USSR, Physics of the Solid Earth (English Translation), **18**: 358–364.

BAZHENOV, M. L., and BURTMAN, V. S. 1982. The kinematics of the Pamir Arc. Geotectonics (English Translation), **16**: 288–301.

——— 1986. Tectonics and paleomagnetism of structural arcs of the Pamir–Punjab syntaxes. Journal of Geodynamics, **5**: 383–396.

BECK, M. E., JR. 1980. Paleomagnetic record of plate-margin tectonic processes along the western edge of North America. Journal of Geophysical Research, **85**: 7115–7131.

CAREY, S. W. 1958. A tectonic approach to continental drift. *In* Continental Drift: a Symposium. *Edited by* S. W. Carey. University of Tasmania, Hobart, Australia, pp. 17–335.

DANKERS, P. H. M. 1978. Magnetic properties of dispersed natural iron-oxides of known grain size. Ph.D. thesis, University of Utrecht, Utrecht, The Netherlands.

ELDREDGE, S., BACHTADSE, V., and VAN DER VOO, R. 1985. Paleomagnetism and the orocline hypothesis. Paleomagnetism and the orocline hypothesis. *In* Collision tectonics: deformation of continental lithosphere. *Edited by* N. L. Carter and S. Uyeda. Tectonophysics, **119**: 153–179.

ENGEBRETSON, D. C., and BECK, M. E., JR. 1978. On the shape of directional data sets. Journal of Geophysical Research, **83**: 5979–5982.

FISHER, R. A. 1953. Dispersion on a sphere. Proceedings of the Royal Astronomical Society of London, Series A: Mathematical and Physical Sciences, **217**: 295–305.

GRUBBS, K. L., and VAN DER VOO, R. 1976. Structural deformation of the Idaho–Wyoming overthrust belt (U.S.A.), as determined by Triassic paleomagnetism. Tectonophysics, **33**: 321–336.

HALLS, H. C. 1976. A least-squares method to find a remanence direction from converging remagnetization circles. Geophysical Journal of the Royal Astronomical Society, **45**: 297–304.

KING, P. B., *compiler*. 1969. Tectonic map of North America. United States Geological Survey, map, scale 1 : 5 000 000.

KNOWLES, R. R., and OPDYKE, N. D. 1968. Paleomagnetic results from the Mauch Chunk Formation: a test of the origin of curvature in the folded Appalachians of Pennsylvania. Journal of Geophysical Research, **73**: 6515–6526.

KOTÁSEK, J., and KRS, M. 1965. Paleomagnetic study of tectonic rotations in the Carpathian Mountains of Czechoslovakia. Palaeogeography, Palaeoclimatology, Palaeoecology, **1**: 39–49.

LAROCHELLE, A., and BLACK, R. F. 1965. The design and testing of an alternating-field demagnetizing apparatus. Canadian Journal of Earth Sciences, **2**: 684–696.

LAROCHELLE, A., and CHRISTIE, K. 1967. Automatic three-magnet or biastatic magnetometer. Geological Survey of Canada, Paper 67–28.

LOWRIE, W., and HIRT, A. M. 1986. Paleomagnetism in arcuate mountain belts. *In* The origin of arcs. *Edited by* F.-C. Wezel. Developments in Geotectonics, **21**: 141–158.

MARSHAK, S. 1988. Kinematics of orocline and arc formation in thin-skinned orogens. Tectonics, **7**: 73–86.

MCELHINNY, M. W. 1964. Statistical significance of the fold test in palaeomagnetism. Geophysical Journal of the Royal Astronomical Society, **8**: 338–340.

MCFADDEN, P. L., and JONES, D. L. 1981. The fold test in palaeomagnetism. Geophysical Journal of the Royal Astronomical Society, **67**: 53–58.

MCFADDEN, P. L., and LOWES, F. J. 1981. The discrimination of mean directions drawn from Fisher distributions. Geophysical Journal of the Royal Astronomical Society, **67**: 19–33.

NORRIS, D. K. 1968. The Crowsnest Deflection of the eastern Cordillera of Canada. Geological Society of America, Annual Meeting, Abstracts with Programs, p. 221.

——— 1972. *En échelon* folding in the northern Cordillera of Canada. Bulletin of Canadian Petroleum Geology, **20**: 634–642.

NORRIS, D. K., and BLACK, R. F. 1961. Application of palaeomagnetism to thrust mechanics. Nature (London), **192**: 933–935.

——— 1962. Palaeomagnetism and differential rotation in the Lewis thrust plate. Journal of the Alberta Society of Petroleum Geologists, **10**: 13–21.

NORRIS, D. K., and YORATH, C. J. 1981. The North American plate from the Arctic Archipelago to the Romanzof Mountains. *In* The ocean basins and margins. Vol. 5. The Arctic Ocean. *Edited by* A. E. M. Nairn, M. Churkin, Jr., and F. G. Stehli. Plenum Press, New York, NY, pp. 37–103.

ÖZDEMIR, O., and BANERJEE, S. K. 1984. High temperature stability of maghemite (γ-Fe_2O_3). Geophysical Research Letters, **11**: 161–164.

ÖZDEMIR, O., and DUNLOP, D. J. 1988. Crystallization remanent magnetization during the transformation of maghemite to hematite. Journal of Geophysical Research, **93**: 6530–6544.

PARK, J. K. 1981. Paleomagnetism of the Late Proterozoic sills in the Tsezotene Formation, Mackenzie Mountains, Northwest Territories, Canada. Canadian Journal of Earth Sciences, **18**: 1572–1580.

ROY, J. L., SANDERS, E., and REYNOLDS, J. 1972. Un four électrique pour l'étude des propriétés magnétiques des roches. Publication de la Division de la Physique du Globe, No. 42, pp. 229–237.

ROY, J. L., REYNOLDS, J., and SANDERS, E. 1973. An alternating field demagnetizer for rock magnetism studies. Energy, Mines and Resources Canada, Publications of the Earth Physics Branch, No. 44, Part 3, pp. 37–45.

SCHWARTZ, S. Y., and VAN DER VOO, R. 1983. Paleomagnetic evaluation of the orocline hypothesis in the central and southern Appalachians. Geophysical Research Letters, **10**: 505–508.

——— 1984. Paleomagnetic study of thrust sheet rotation during foreland impingement in the Wyoming–Idaho overthrust belt. Journal of Geophysical Research, **89**: 10 077 – 10 086.

STOCKWELL, C. H. 1969. Tectonic map of Canada. Geological Survey of Canada, Map 1251A, scale 1 : 5 000 000.

WILSON, J. T. 1954. The development of structure of the crust. *In* The Earth as a planet of the Solar System. *Edited by* G. P. Kuipes. University of Chicago Press, Chicago, IL, pp. 138–214.